

## Practical – 1

**AIM:** Study of various type of Software Process models with comparison and find out which process model will be appropriate for Resume Skill Gap Analyzer.

**Objectives:** To learn different process models and identify suitable model for the project development.

- **Theory:**

A software process is defined as a collection of work activities, actions, and tasks that are performed when some work product is to be created.

**List of Different Process Models**

- Waterfall model.
- V model.
- Incremental model.
- RAD model.
- Agile model.
- Iterative model.
- Spiral model.
- Prototype model.
- Agile model

**Quiz:**

1. Compare the waterfall model and incremental model.

| Waterfall Model                                                                                                                  | Incremental Model                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| The Waterfall model follows a linear and sequential approach where each phase must be completed before moving to the next phase. | The Incremental model develops the software in small parts called increments, and each increment adds new functionality. |
| Customer feedback is obtained only after the complete software is developed.                                                     | Customer feedback is obtained after every increment of development.                                                      |
| The Waterfall model is rigid and does not easily allow changes once development has started.                                     | The Incremental model is flexible and allows changes during development.                                                 |
| Testing is performed only after the implementation phase is completed.                                                           | Testing is performed after each increment is developed.                                                                  |

| Waterfall Model                                               | Incremental Model                                                    |
|---------------------------------------------------------------|----------------------------------------------------------------------|
| Risk is high because the entire product is delivered at once. | Risk is reduced because the product is delivered in parts.           |
| The complete product is delivered at the end of the process.  | A working version of the software is delivered after each increment. |

2. State whether the following statements are true or false. Justify your answer.

- a) Software development organizations which follow the iterative waterfall model for product development provide maximum customer satisfaction.

**Ans:** false, because the iterative waterfall model still restricts customer involvement until later stages of development. Since customer feedback is not continuously incorporated throughout the development process, the final product may not fully meet customer expectations. Models such as Agile and Incremental provide higher customer satisfaction because they involve users frequently and allow changes based on feedback.

- b) The loops for the spiral model are fixed.

**Ans:** false, because in the Spiral model, the number of loops is not fixed. Each loop of the spiral represents a phase of the software development process and is determined by the size, complexity, and risk level of the project. The loops may increase or decrease depending on project requirements and risk analysis.

### Suggested Reference:

1. Ian Sommerville, Software engineering, Pearson education Asia
2. Roger S.Pressman, Software Engineering- A practitioner's Approach, McGraw-Hill International Editions

### References used by the students:

### Rubric wise marks obtained:

| Rubrics | 1 | 2 | 3 | 4 | 5 | Total |
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| Marks | Complete implementation as asked | Complete implementation as asked<br><br>Problem analysis | Complete implementation as asked<br><br>Problem analysis<br><br>Development of the Solution | Complete implementation as asked<br><br>Problem analysis<br><br>Development of the Solution<br><br>Concept Clarity & understanding | Complete implementation as asked<br><br>Problem analysis<br><br>Development of the Solution<br><br>Concept Clarity & understanding<br><br>Correct answer to all questions |  |
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**Signature of Faculty:**